

Python tutorial Part 2
Data analysis and visualization with Python
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COMM2708 Digital and Data Journalism

Requirements

1. Jupyter notebook
2. Dataset: DDJ_3_Dataset_Incidents.xlsx put in the same folder with your Python file
3. Required packages: pandas, matplotlib, and openpyxl

PART 1 Explore the Dataset

1. Preview the first 10 rows of the dataset
2. Display the list of column names
3. Identify the columns you will use for year, offence categories, Police Service Area and incidents counts

Part 2 Grouping, Summation, and Sorting

1. Group the data by Office Divisions and calculate the total Incidents Recorded for each division
2. Group the data by Police Service Area and calculate the total Incidents Recorded for each area.

Answer these questions:

1. What are the top three offence divisions with the highest number of incidents recorded?
2. What are the three offence divisions with the lowest number of incidents recorded?
3. What are the top three Police Service Areas with the highest number of incidents recorded?
4. What are the three Police Service Areas with the lowest number of incidents recorded?

Part 3 Filtering 1

1. Filter the datasets only with Crimes against the person
2. Display all unique categories in offence division
3. Filter the dataset so it contains only a Crimes against the person
4. Display the unique Offence Subdivision categories within Crimes against the person

Part 4 Filtering 2

1. Filter the datasets only with Stalking, Harassment, and Threatening Behaviour

Answer these questions:

1. How many stalking, harassment, and threatening behaviour incidents were recorded in total?
2. How many stalking, harassment, and threatening behaviour incidents were recorded in each year?
3. What are the top three years with the most stalking, harassment, and threatening behaviour incidents?

4. What are the three years with the fewest stalking, harassment, and threatening behaviour incidents?
5. What are the top three Police Service Areas with the most FV stalking incidents?
6. What are the top three Police Service Areas with the most Non FV stalking incidents?

Part 5 Data Visualizations

1. Visualization 1: All incidents by year (Line graph)
2. Visualization 2: Stalking, harassment, and threatening behaviour by year (Line graph)
3. Visualization 3: Police service areas (Bar graph)